



THE UNIVERSITY OF HARIPUR

Department of Information Technology • www.uoh.edu.pk

WEEK 8 INTERNSHIP PROGRESS REPORT

Remote Internship at FlyRank AI

STUDENT INFORMATION

University Name	University of Haripur
Student Name	Muhammad Abdullah Awan
Roll Number	F23-0332
Department	Department of Information Technology
Program	BS Artificial Intelligence
Semester	7th Semester
Company	FlyRank AI
Internship Type	Remote Internship
Internship Duration	July 2026 – August 2026 (8 Weeks)
Report Period	Week 8 (August 19 – August 25, 2026)

Submitted To:

Dr. Dawar Khan

Submitted By:

Muhammad Abdullah Awan

1. Introduction

FlyRank AI is an AI-powered organic growth platform that helps businesses improve their visibility across traditional search engines and AI answer engines such as ChatGPT, Perplexity, and Gemini. The platform automates large parts of the content lifecycle - research, scoring, and optimization - primarily serving D2C, SaaS, and agency clients who need scalable organic growth.

This report covers Week 8 - the final capstone week - of my remote Machine Learning Engineering internship at FlyRank AI, under the Applied Search Intelligence track. It summarizes the capstone deliverable completed, the reasoning behind it, and the skills developed.

2. Objective of Week 8

Objective: To assemble every weekly assignment into one deployed, public research paper built on real FlyRank search data - the final capstone deliverable of the Machine Learning Engineering track (ML-CAP-01) - and to tie the findings back to FlyRank's real content-refresh problem in public-safe language.

3. Week 8: Capstone: Deployed Research Paper

Task Requirements: Build a capstone notebook mirroring the paper's structure (Question, Data, Methodology, Results vs. baseline, Limitations, Ranked Recommendations, Artifacts); write and deploy a full public research paper containing all required sections (Abstract, Introduction, Data, Methodology, Results with charts, Limitations, Recommendations, Reproducibility, Acknowledgments and data credit); deploy the paper to a public URL and record it in submission/paper_url.txt; strengthen the paper's case-study framing; add a 5-minute demo outline and two shareable cuts of the work.

Research Performed: Spent time reviewing how well-structured public research pages are built (canonical section order, chart hygiene, honest-claim language) before writing, per this week's session guidance, and studied FlyRank's own published research paper as a structural reference for tone and section design.

Summary of Work Completed: Assembled capstone.ipynb, consolidating the question, data sources, methodology, and honest client-holdout results from Weeks 4-7 into one coherent notebook. Wrote a full research paper as a static webpage ('Which Pages Should You Refresh First?') containing all nine required sections, including a 5-sentence abstract, an honest Results section reporting the Random Forest model's measured Precision@50 of 0.540 against the baseline rule's 0.240 (a ~2.25x lift) on an identical client-holdout split, a Limitations section written in observed/directional/decision-support language, a ranked action-recommendation playbook, and an Acknowledgments section crediting the FlyRank ML Internship dataset with a link to flyrank.ai. Deployed the paper via GitHub Pages directly from the repository's /docs folder, debugged relative image paths so all charts render correctly on the live public page, and recorded the exact deployed URL in submission/paper_url.txt as required. As a final polish task, added explicit case-study framing connecting the work to FlyRank's own research on refresh timing, a 5-minute demo outline for the optional showcase, and two shareable cuts (a social post and a 3-sentence employer-facing summary) committed to the closing section of the capstone notebook.

What Was Learned: Learned that turning eight weeks of separate technical notebooks into one coherent, honestly-told public narrative is a distinct skill from the analysis itself - structure, an honest Limitations section, and clear storytelling carry as much weight as the underlying numbers. Also learned that GitHub Pages makes deploying a notebook's findings as a real, shareable public artifact genuinely accessible, with no new accounts or hosting costs.

4. Skills Developed

Week 8 contributed to the development of the following technical, analytical, and professional skills:

Skill	Description
Research Communication	Structured and wrote a complete public research paper following the canonical abstract-through-acknowledgments format.
Static Site Deployment	Deployed a working public webpage via GitHub Pages, including debugging relative asset paths.

Skill	Description
Narrative Synthesis	Consolidated eight weeks of separate technical notebooks into one coherent, honestly-told capstone story.
Audience Adaptation	Repurposed the same findings into a 5-minute demo outline, a social post, and a 3-sentence employer-facing summary.

5. Challenges Faced

- › Debugging GitHub Pages image rendering - charts initially failed to display on the live page due to incorrect relative paths, requiring a careful review of the served folder structure to fix.
- › Condensing eight weeks of technical work into a 5-sentence abstract and a 5-minute demo outline without losing the honest, validated result required several rewrites to stay both concise and accurate.

6. Knowledge and Experience Gained

Week 8 provided practical, end-to-end experience in the last mile of applied ML work that most coursework skips entirely: turning a validated result into a public, deployed, honestly-told research artifact that a stranger - an interviewer, a recruiter, or the wider community - can read and trust in minutes.

7. Conclusion

Week 8 completed the FlyRank AI Machine Learning Engineering internship capstone: a full research paper, built on real production-scale search data, honestly validated against a transparent baseline, and deployed to a public URL with a fully reproducible repository behind it.

This marks the completion of the core ML track deliverables (5+ weekly assignments and the capstone). The optional General AI Fluency assignments and further refinement of the deployed paper remain available as stretch goals going forward.